for the harmonic one, then we encounter more complicated *nonlinear resonances*. It is challenging (perhaps impossible) to analyze the destabilizing effect of such resonances in a general molecular dynamics setting, but it is possible to do so in some cases for low-dimensional model problems [247, 329, 339].

In practice the presence of resonances may or may not render the multiple timestepping technique ineffective. One would be concerned if the only gain were from the Verlet stability threshold at $\bar{h}_{\text{Verlet}} = 2/\Omega$ to the location of the first resonance peak at $\bar{h}_{\text{MTS}} = \pi/\Omega$, since the multiple timestepping method introduces additional complexity. However, the picture is complicated by the observation that the strength of the leading resonance is often modest, so the growth observed in practice may be slow and, depending on the nature of the coupling of the variables of the system (which depends on details of the potential energy functions, natural coupling of the modes of the system, and the presence of solvent molecules which may introduce a stochastic effect) the growth of error due to resonance may be masked by other factors of a strongly system-dependent nature. Moreover, molecular systems are often subject to a stochastic thermostat (a topic to be addressed in later chapters) which may further alter the role of the resonances. Thus the study of the effect of linear and nonlinear resonance peaks on the error in molecular dynamics simulation becomes quickly a numerical issue which must be investigated using carefully crafted experiments. The analysis of [245] clearly raises some concerns regarding the use of multiple timestepping in molecular simulation, in particularly they suggest that the usable gain from RESPA simulations in systems with detailed solvent (water) is only around a factor of $\pi/2$ compared what can be achieved using a standard integrator such as Verlet (similar what would be suggested by resonance analysis). Nevertheless there are practical examples where multiple timestepping is used, especially in schemes incorporating stochastic perturbation and with relatively strong scale separation, as when multiple timestepping is combined with the particle-mesh Ewald summation long-ranged force computation technique. In Sect. 7.13 we discuss the use of multiple timestepping in connection with Langevin dynamics. In Sect. 8.6, we describe the combination of multiple timestepping with isokinetic constraints and stochastic perturbation, where very substantial increases in outer timestep are found to be possible.

### *4.2.2 The Mollified Impulse Method*

The onset of instability of multiple timestepping is a consequence of introducing periodic impulses (via slow force updates) in resonance with a natural frequency of the system. As a means of avoiding or at least ameliorating the effect of resonances, it has been proposed to modify the points at which the slow forces in multiple timestepping are evaluated. The *mollified impulse method* (MOLLY) [141] is a form of multiple timestepping in which the potential energy function is modified based on an averaging of the position variable (over fast dynamics) in such a way as to filter out destabilizing excitations. Effectively, one replaces $U_{\text{S}}(\boldsymbol{q})$ wherever it appears

within the MTS method by a modified potential function $\bar{U}(\boldsymbol{q}) \stackrel{\text{def}}{=} U_{\text{S}}(\mathcal{A}(\boldsymbol{q}))$. The mapping $\mathcal{A}(\boldsymbol{q})$ is defined by reference to the fast dynamics, e.g. by the weighted average

$$\mathcal{A}(\boldsymbol{q}) = \frac{1}{K+1} \sum_{i=0}^{K} \phi_i \boldsymbol{\theta}_i(\boldsymbol{q}),$$

where the $\boldsymbol{\theta}_i$ are defined recursively by solving the system with Hamiltonian

$$H^{\text{fast}}(\boldsymbol{q}, \boldsymbol{p}) = \boldsymbol{p}^T \boldsymbol{M}^{-1} \boldsymbol{p}/2 + U_{\text{F}}(\boldsymbol{q})$$

starting from initial position $\boldsymbol{q}$ and zero initial momentum $\boldsymbol{p}$, for $K$ timesteps using a small stepsize $\tau_f$. This small step does not have to be directly related to the timestep used for integrating the fast dynamics within the MTS scheme; it is only used for the averaging procedure. The resulting sequence of $\boldsymbol{\theta}_i$ clearly are functions of $\boldsymbol{q}$ only and so $\mathcal{A}$ is well defined. The weights $\phi_i$ appearing in the average may be selected in various ways. The initialization of the calculation of the $\boldsymbol{\theta}_i$ can, moreover, be modified by various alternative $\boldsymbol{q}$-dependent proposals, or may be randomized.

An alternative, simpler, and probably more robust, procedure is simply to fix the value of $\mathcal{A}(\boldsymbol{q})$ by constraining $\boldsymbol{q}$ to its mechanical equilibrium value [189] in which case no fast averaging whatsoever needs to be performed. This approach relies on the use of the SHAKE method of constrained molecular dynamics (see the following section).

In each of the mollified multiple timestepping methods, since only the potential energy is modified, and by a smooth mapping of positions, the forces derived by differentiating $\bar{U}$ are still conservative, and, just as with the MTS method, the scheme remains symplectic.

The mollification is only applied for the purpose of modifying the slow impulse and the mollified methods are in other ways identical to multiple timestepping. These schemes appear to confer some distinct advantage in the large stepsize in molecular dynamics. For example where a detailed and fully flexible molecular model of a solvated biomolecule might require stepsizes of 2 fs in a Verlet simulation, mollified impulse methods may allow increases to 6 fs [189] in the 'large' timestep, i.e. the period between reevaluations of the slow forces. This benefit may be substantial in simulations of large systems.

## 4.3 Methods for Constrained Dynamics

Constraints are used frequently in molecular modelling. By a constraint, we mean a modification of a dynamical system to maintain the constancy of a given function of coordinates (or coordinates and momenta). In this section we consider *holonomic* constraints which may be expressed in terms of positions only, and perhaps time,